4.4.2 Properties of the cosine function

From the graph of $y = \cos x$, we observe the following properties of cosine function:

(i) There is no break or discontinuities in the curve. The cosine function is continuous.

(ii) The cosine function is even, since the graph is symmetric about y-axis.

(iii) The maximum value of cosine function is 1 and occurs at $x = \ldots, -2\pi, 0, 2\pi, \ldots$ and the minimum value is – 1 and occurs at $x = \ldots, -\pi, \pi, 3\pi, 5\pi, \ldots$. In other words, $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$.

Remark

(i) Shifting the graph of $y = \cos x$ to the right $\frac{\pi}{2}$ radians, gives the graph of $y = \cos\left(x - \frac{\pi}{2}\right)$, which is same as the graph of $y = \sin x$. Observe that $\cos\left(x - \frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2} - x\right) = \sin x$.

(ii) $y = A \sin \alpha x$ and $y = B \cos \beta x$ always satisfy the inequalities $-|A| \leq A \sin \alpha x \leq |A|$ and $-|B| \leq B \cos \beta x \leq |B|$. The amplitude and period of $y = \mathrm{A} \sin \alpha x$ are $|A|$ and $\frac{2\pi}{|\alpha|}$, respectively and those of $y = B \cos \beta x$ are $|B|$ and $\frac{2\pi}{|\beta|}$, respectively.

The functions $y = A \sin \alpha x$ and $y = B \cos \beta x$ are known as **sinusoidal functions**.

(iii) Graphing of $y = A \sin \alpha x$ and $y = B \cos \beta x$ are obtained by extending the portion of the graphs on the intervals $\left[0, \frac{2\pi}{|\alpha|}\right]$ and $\left[0, \frac{2\pi}{|\beta|}\right]$, respectively.



Applications

Phenomena in nature like tides and yearly temperature that cycle repetitively through time are often modelled using **sinusoids**. For instance, to model tides using a general form of sinusoidal function $y = d + a\cos(bt - c)$, we give the following steps:

(i) The amplitude of a sinusoidal graph (function) is one-half of the absolute value of the difference of the maximum and minimum y-values of the graph.

Thus, **Amplitude**, $a = \frac{1}{2}(\text{max} - \text{min})$; Centre line is $y = d$, where $d = \frac{1}{2}(\text{max} + \text{min})$

(ii) **Period**, $p = 2 \times$ (time from maximum to minimum) ; $b = \frac{2\pi}{p}$

(iii) $c = b \times$ time at which maximum occurs.

Model-1

The depth of water at the end of a dock varies with tides. The following table shows the depth (in metres) of water at various time.

time, t	12 am	2 am	4 am	6 am	8 am	10 am	12 noon
depth	3.5	4.2	3.5	2.1	1.4	2.1	3.5

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